

WORKED EXAMPLE

Keep the model simple to isolate batching

One trainable number predicts the same output for every example.

$$\hat{y}_i = w, \quad (t_1, t_2, t_3, t_4) = (0, 2, 4, 6)$$

$$\ell_i(w) = \frac{1}{2}(w - t_i)^2, \quad J(w) = \frac{1}{4} \sum_{i=1}^4 \ell_i(w)$$

$$\frac{d\ell_i}{dw} = \frac{1}{2} \cdot 2(w - t_i) \cdot 1 = w - t_i$$

$$w_0 = 0, \quad \eta = 0.1$$

This constant-output model intentionally removes network details so we can see when gradients are evaluated and when parameters change.

Use the same data and initial parameter for each method. The learning rate is fixed here to isolate batching; lecture algorithms can also schedule it.

WORKED EXAMPLE

Examples, batches, updates and epochs

An epoch is one pass through all training examples.

An example is one data item with a target. Its gradient is computed at the current parameter. An update changes the parameter using a gradient or an average of gradients.

Full batch: evaluate all four example gradients at the same old w , average once, then update once.

SGD here means batch size one: evaluate one example gradient, update immediately, then evaluate the next example at the changed parameter.

Mini-batch: average gradients from a small group at one common w , update once, then use the new w for the next group.

$$g_B(w) = \frac{1}{|B|} \sum_{i \in B} (w - t_i), \quad w_{new} = w - \eta g_B(w)$$

B is the set of example indices in this batch; $|B|$ is its number of examples. Our chosen orders are fixed for reproducibility; the lecture shuffles examples.

WORKED EXAMPLE

Full batch, update 1

Reset to the original start only when beginning a different method.

$$w_{old} = 0, \quad \text{targets} = (0, 2, 4, 6)$$

$$g_{t=0} = 0 - 0 = 0$$

$$g_{t=2} = 0 - 2 = -2$$

$$g_{t=4} = 0 - 4 = -4$$

$$g_{t=6} = 0 - 6 = -6$$

$$\bar{g} = \frac{(0) + (-2) + (-4) + (-6)}{4} = -3$$

$$w_{new} = 0 - 0.1(-3) = 0.3$$

Every gradient above uses exactly the same old parameter. Only after averaging do we change w .

WORKED EXAMPLE

SGD forward, update 1

Reset to the original start only when beginning a different method.

$$w_{old} = 0, \quad \text{targets} = (0)$$

$$g_{t=0} = 0 - 0 = 0$$

$$w_{new} = 0 - 0.1(0) = 0$$

The next example will use this newly updated parameter. Do not return to the original w when computing its gradient.

WORKED EXAMPLE

SGD forward, update 2

Reset to the original start only when beginning a different method.

$$w_{old} = 0, \quad \text{targets} = (2)$$

$$g_{t=2} = 0 - 2 = -2$$

$$w_{new} = 0 - 0.1(-2) = 0.2$$

The next example will use this newly updated parameter. Do not return to the original w when computing its gradient.

WORKED EXAMPLE

SGD forward, update 3

Reset to the original start only when beginning a different method.

$$w_{old} = 0.2, \quad \text{targets} = (4)$$

$$g_{t=4} = 0.2 - 4 = -3.8$$

$$w_{new} = 0.2 - 0.1(-3.8) = 0.58$$

The next example will use this newly updated parameter. Do not return to the original w when computing its gradient.

WORKED EXAMPLE

SGD forward, update 4

Reset to the original start only when beginning a different method.

$$w_{old} = 0.58, \quad \text{targets} = (6)$$

$$g_{t=6} = 0.58 - 6 = -5.42$$

$$w_{new} = 0.58 - 0.1(-5.42) = 1.122$$

The next example will use this newly updated parameter. Do not return to the original w when computing its gradient.

WORKED EXAMPLE

SGD reverse, update 1

Reset to the original start only when beginning a different method.

$$w_{old} = 0, \quad \text{targets} = (6)$$

$$g_{t=6} = 0 - 6 = -6$$

$$w_{new} = 0 - 0.1(-6) = 0.6$$

The next example will use this newly updated parameter. Do not return to the original w when computing its gradient.

WORKED EXAMPLE

SGD reverse, update 2

Reset to the original start only when beginning a different method.

$$w_{old} = 0.6, \quad \text{targets} = (4)$$

$$g_{t=4} = 0.6 - 4 = -3.4$$

$$w_{new} = 0.6 - 0.1(-3.4) = 0.94$$

The next example will use this newly updated parameter. Do not return to the original w when computing its gradient.

WORKED EXAMPLE

SGD reverse, update 3

Reset to the original start only when beginning a different method.

$$w_{old} = 0.94, \quad \text{targets} = (2)$$

$$g_{t=2} = 0.94 - 2 = -1.06$$

$$w_{new} = 0.94 - 0.1(-1.06) = 1.046$$

The next example will use this newly updated parameter. Do not return to the original w when computing its gradient.

WORKED EXAMPLE

SGD reverse, update 4

Reset to the original start only when beginning a different method.

$$w_{old} = 1.046, \quad \text{targets} = (0)$$

$$g_{t=0} = 1.046 - 0 = 1.046$$

$$w_{new} = 1.046 - 0.1(1.046) = 0.9414$$

The next example will use this newly updated parameter. Do not return to the original w when computing its gradient.

WORKED EXAMPLE

Mini-batch, update 1

Reset to the original start only when beginning a different method.

$$w_{old} = 0, \quad \text{targets} = (0, 2)$$

$$g_{t=0} = 0 - 0 = 0$$

$$g_{t=2} = 0 - 2 = -2$$

$$\bar{g} = \frac{(0) + (-2)}{2} = -1$$

$$w_{new} = 0 - 0.1(-1) = 0.1$$

Every gradient above uses exactly the same old parameter. Only after averaging do we change w .

WORKED EXAMPLE

Mini-batch, update 2

Reset to the original start only when beginning a different method.

$$w_{old} = 0.1, \quad \text{targets} = (4, 6)$$

$$g_{t=4} = 0.1 - 4 = -3.9$$

$$g_{t=6} = 0.1 - 6 = -5.9$$

$$\bar{g} = \frac{(-3.9) + (-5.9)}{2} = -4.9$$

$$w_{new} = 0.1 - 0.1(-4.9) = 0.59$$

Every gradient above uses exactly the same old parameter. Only after averaging do we change w .

WORKED EXAMPLE

Same epoch, different sequences of parameter states

Each method evaluates four example gradients in this one epoch.

method	parameter after each update	updates
Full batch	0.3	1
SGD forward	0, 0.2, 0.58, 1.122	4
SGD reverse	0.6, 0.94, 1.046, 0.9414	4
Mini-batch	0.1, 0.59	2

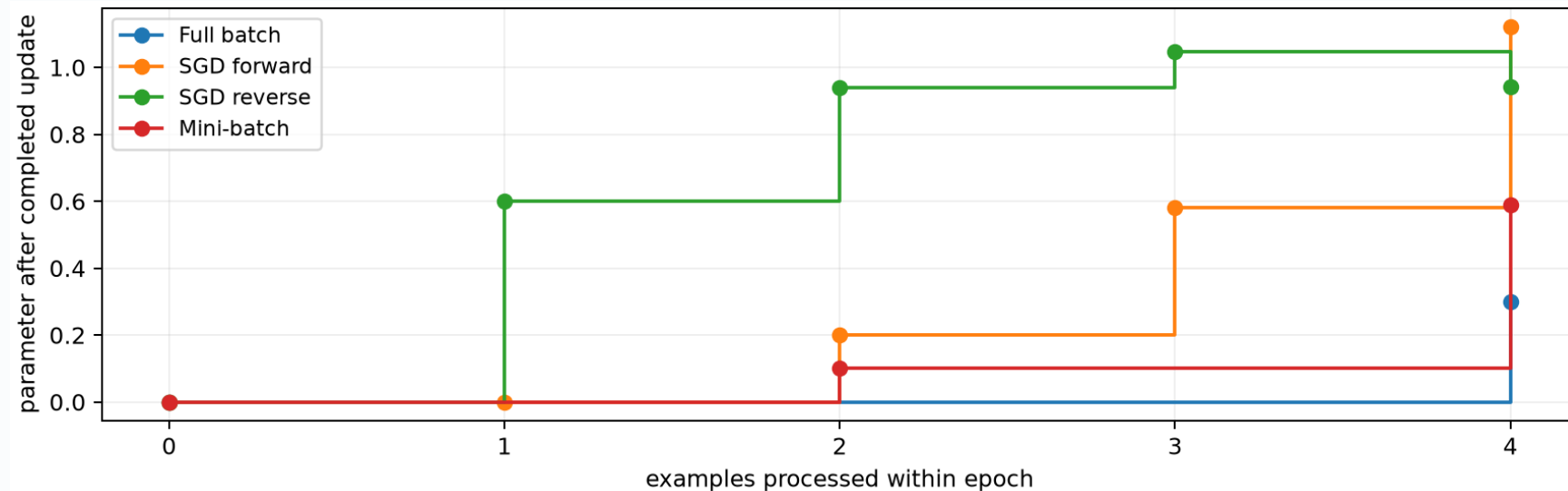
Full batch evaluates all four gradients at $w=0$. SGD changes w between examples, which is why reversing the order changes the result. Mini-batches change w only between groups.

The comparison uses equal one-epoch example-gradient counts, but different numbers of updates. It is not an equal-runtime claim, and the same rate is not guaranteed to be optimal for every method.

WORKED EXAMPLE

Compare paths by examples processed

Each curve advances through the same four training examples.



Full batch first changes w after four examples. Mini-batches change it after two and four examples. SGD changes it after each example; the zero first forward-order gradient leaves w unchanged.

WORKED EXAMPLE

Report the full objective at the final parameter

All reported J values use all four targets, even for SGD or mini-batch training.

$$J(0.3) = \frac{(0.3 - 0)^2 + (0.3 - 2)^2 + (0.3 - 4)^2 + (0.3 - 6)^2}{8}$$
$$= \frac{0.09 + 2.89 + 13.69 + 32.49}{8} = 6.145$$

The divisor is 8 because each loss has factor 1/2 and the objective averages four losses.

method	final w	J on all four examples
Full batch	0.3	6.14500000
SGD forward	1.122	4.26344200
SGD reverse	0.9414	4.61891698
Mini-batch	0.59	5.40405000

These values describe this fixed-rate, one-epoch example. They do not prove a universal ranking of the methods.

WORKED EXAMPLE

Count updates and retain the last partial batch

Batch size is 20, but the dataset has 103 examples.

$$103 = 5(20) + 3 \implies \text{sizes : } 20, 20, 20, 20, 20, 3$$

$$\text{updates/epoch} = \lceil 103/20 \rceil = 6$$

$$20 \text{ epochs : } \text{mini} = 20(6) = 120$$

$$\text{SGD} = 20(103) = 2060, \quad \text{full batch} = 20(1) = 20$$

For the last mini-batch gradient, divide its gradient sum by its actual size 3. That divisor is distinct from the 103-example mean used to report the full dataset objective.

The ceiling brackets mean round up to the next integer. We retain the partial batch; dropping it would change the update count.

INDEPENDENT PRACTICE

Your turn: fresh targets and a different rate

Restart each method from $w_0=0$.

$$(t_1, t_2) = (1, 3), \quad \eta = 0.2, \quad \hat{y} = w$$

Calculate one full-batch update and one complete SGD epoch in each order: $[1,3]$ and $[3,1]$. Use the same half-squared per-example loss as before.

List every gradient and intermediate parameter. Explain why the full-batch average uses the original w throughout, while SGD does not.

Separately, retain all examples in a dataset of size 10 with batch size 4. Give each batch size, updates per epoch, and the divisor of the last gradient mean.

WORKED ANSWER

Practice: Full batch, update 1

Reset to the original start only when beginning a different method.

$$w_{old} = 0, \quad \text{targets} = (1, 3)$$

$$g_{t=1} = 0 - 1 = -1$$

$$g_{t=3} = 0 - 3 = -3$$

$$\bar{g} = \frac{(-1) + (-3)}{2} = -2$$

$$w_{new} = 0 - 0.2(-2) = 0.4$$

Every gradient above uses exactly the same old parameter. Only after averaging do we change w .

WORKED ANSWER

Practice: SGD forward, update 1

Reset to the original start only when beginning a different method.

$$w_{old} = 0, \quad \text{targets} = (1)$$

$$g_{t=1} = 0 - 1 = -1$$

$$w_{new} = 0 - 0.2(-1) = 0.2$$

The next example will use this newly updated parameter. Do not return to the original w when computing its gradient.

WORKED ANSWER

Practice: SGD forward, update 2

Reset to the original start only when beginning a different method.

$$w_{old} = 0.2, \quad \text{targets} = (3)$$

$$g_{t=3} = 0.2 - 3 = -2.8$$

$$w_{new} = 0.2 - 0.2(-2.8) = 0.76$$

The next example will use this newly updated parameter. Do not return to the original w when computing its gradient.

WORKED ANSWER

Practice: SGD reverse, update 1

Reset to the original start only when beginning a different method.

$$w_{old} = 0, \quad \text{targets} = (3)$$

$$g_{t=3} = 0 - 3 = -3$$

$$w_{new} = 0 - 0.2(-3) = 0.6$$

The next example will use this newly updated parameter. Do not return to the original w when computing its gradient.

WORKED ANSWER

Practice: SGD reverse, update 2

Reset to the original start only when beginning a different method.

$$w_{old} = 0.6, \quad \text{targets} = (1)$$

$$g_{t=1} = 0.6 - 1 = -0.4$$

$$w_{new} = 0.6 - 0.2(-0.4) = 0.68$$

The next example will use this newly updated parameter. Do not return to the original w when computing its gradient.

WORKED ANSWER

Practice checkpoint and remainder calculation

Different orders can produce different parameter states after one epoch.

$$w_{full} = 0.4, \quad w_{forward} = 0.76, \quad w_{reverse} = 0.68$$

$$10 = 2(4) + 2 \implies \text{batch sizes} = 4, 4, 2$$

$$\text{updates/epoch} = 3, \quad g_{last} = \frac{g_9 + g_{10}}{2}$$

The last gradient mean divides by 2, not by 4 or 10. A full-dataset mean loss would instead average all ten example losses.

Common errors: changing w partway through a full-batch average, averaging gradients from different parameter states, using the nominal size for a shorter last batch, or confusing an epoch with an update.